

Optimization

4.3.2 Barrier / interior penalty function method (IPFM)

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We want a function $B(\mathbf{x})$ that is small in the interior ($g(\mathbf{x}) > 0$) and such that

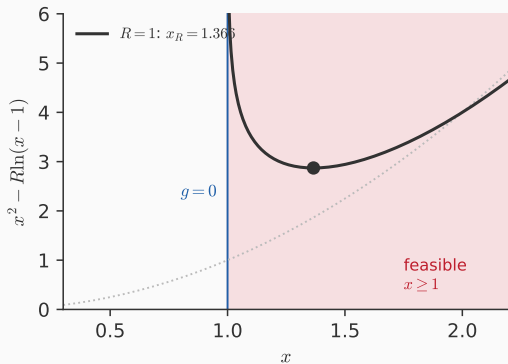
$$B(\mathbf{x}) \rightarrow +\infty \quad \text{as} \quad g(\mathbf{x}) \rightarrow 0^+.$$

The central choice: **$B(\mathbf{x}) = -\ln g(\mathbf{x})$** .

Idea: a wall before the boundary

$\min x^2$ s.t. $g = x - 1 \geq 0$ — the same as in 4.3.1. Now, only for $x > 1$:

$$P(x, R) = x^2 - R \ln(x - 1).$$

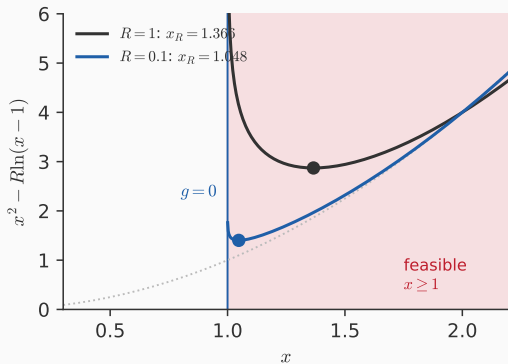


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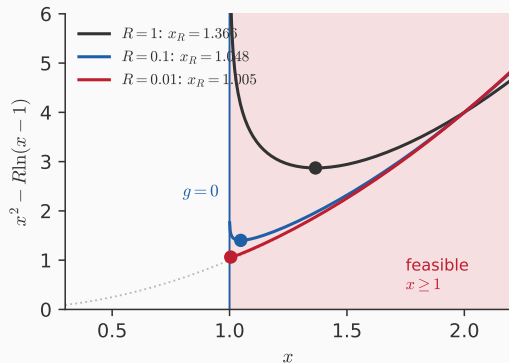
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Question: what happens to the minimum of P as $R \downarrow 0$?

$$2x - \frac{R}{x-1} = 0 \Rightarrow x_R = \frac{1 + \sqrt{1 + 2R}}{2}:$$
$$1.366 \rightarrow 1.048 \rightarrow 1.005 \rightarrow \dots \rightarrow 1^+.$$

The wall moves ever closer to the boundary and the minimum approaches $x^* = 1$ — **always from the feasible side**.



The logarithmic barrier

For $\min f(\mathbf{x})$ s.t. $g_j(\mathbf{x}) \geq 0, j = 1, \dots, J$:

Logarithmic barrier function ($R > 0$)

$$P(\mathbf{x}, R) = f(\mathbf{x}) - R \sum_{j=1}^J \ln g_j(\mathbf{x}), \quad \text{defined only where } g_j(\mathbf{x}) > 0 \text{ for all } j.$$

- Far from the boundary, $-R \ln g_j$ is small for small R : $P \approx f$.
- Near the boundary, $-\ln g_j \rightarrow +\infty$: the wall prevents leaving.
- P is minimized with Ch. 3, for ever smaller R .

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- Near the boundary, $-\ln g_j \rightarrow +\infty$: the wall prevents leaving.
- P is minimized with Ch. 3, for ever smaller R .
- It only works for **inequalities**: **equalities have no interior** — there is no “inside” of $h(\mathbf{x}) = 0$ in which to put a wall. (In practice they are combined: barrier for g_j , penalty $R^{-1}h_\ell^2$ for h_ℓ .)

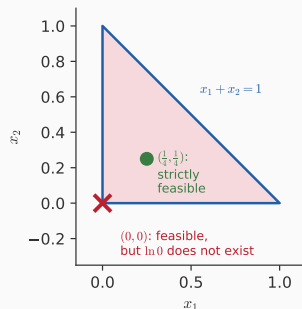
Why do we need an interior point?

$\ln g_j(\mathbf{x})$ only exists if $g_j(\mathbf{x}) > 0$. Hence

$\mathbf{x}^{(0)} \in \text{int } X$ and **all** iterates stay in the interior.

Example: the triangle $x_1 \geq 0$, $x_2 \geq 0$, $1 - x_1 - x_2 \geq 0$. The barrier $-\ln x_1 - \ln x_2 - \ln(1 - x_1 - x_2)$ requires $x_1 > 0$, $x_2 > 0$, $x_1 + x_2 < 1$: $(0, 0)$ is feasible but **will not do**; $(\frac{1}{4}, \frac{1}{4})$ will.

feasible $\neq$ strictly feasible



Advantage

All intermediate solutions are **feasible**: we can stop at any moment.

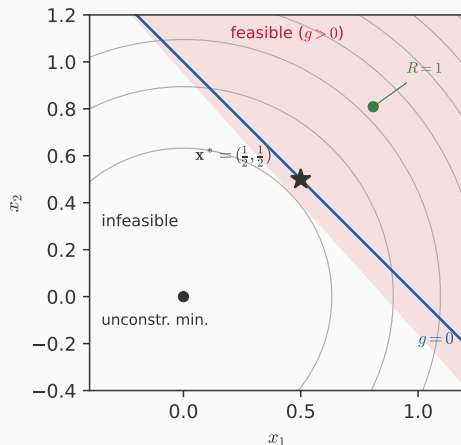
Limitation

We must **find** a strictly feasible point (**Phase I**); the line search cannot cross the boundary.

The same problem as in 4.3.1, now from inside

$\min x_1^2 + x_2^2$ s.t. $g = x_1 + x_2 - 1 \geq 0$; $\mathbf{x}^* = (\frac{1}{2}, \frac{1}{2})$,
 $u^* = 1$.

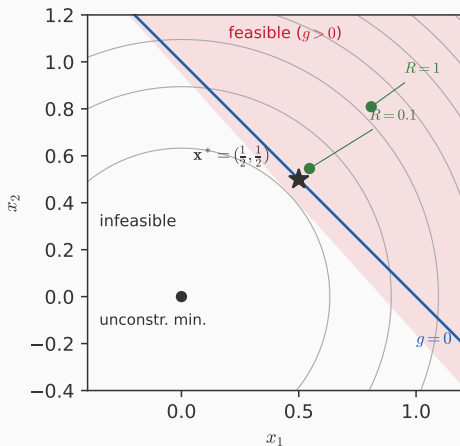
$P = x_1^2 + x_2^2 - R \ln(x_1 + x_2 - 1)$, from an interior
point, for $R = 1, 0.1, 0.01, 0.001$.



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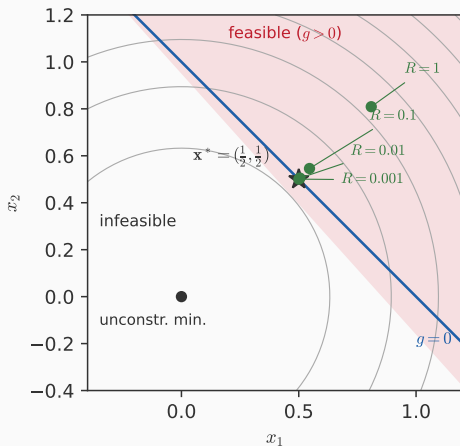
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Question: are the points $\mathbf{x}_R$ feasible?

Yes: all have $g(\mathbf{x}_R) > 0$. They approach $\mathbf{x}^*$ from the **feasible side** — hence the name **interior penalty**.

In 4.3.1 the $\mathbf{x}_R$ came from the unconstrained minimum $(0,0)$, from outside; now they come from the interior of X .



The example worked out

By symmetry $x_1 = x_2 = s$: $\frac{d}{ds} [2s^2 - R \ln(2s - 1)] = 4s - \frac{2R}{2s - 1} = 0 \Rightarrow s_R = \frac{1 + \sqrt{1 + 4R}}{4} > \frac{1}{2}$.

R	$\mathbf{x}_R$	$g(\mathbf{x}_R)$	$\ \mathbf{x}_R - \mathbf{x}^*\ $
1	(0.809, 0.809)	0.618	0.437
0.1	(0.5458, 0.5458)	0.0916	0.0648
0.01	(0.5050, 0.5050)	0.0099	0.0070
0.001	(0.5005, 0.5005)	0.0010	0.00071

- **Does it work?** Yes: $\mathbf{x}_R \rightarrow \mathbf{x}^*$, always with $g(\mathbf{x}_R) > 0$.
- **How fast?** $g(\mathbf{x}_R) \approx R$ and $\|\mathbf{x}_R - \mathbf{x}^*\| = g(\mathbf{x}_R)/\sqrt{2} = O(R)$.
- $f(\mathbf{x}_R) > f^*$: from inside, f decreases to f^* .

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Another barrier: R/g (inverse barrier)

$P = f + R \sum_j 1/g_j$. On the same problem, $\|\mathbf{x}_R - \mathbf{x}^*\| = O(\sqrt{R})$: with $R = 0.01$, $\mathbf{x}_R = (0.548, 0.548)$, against $(0.505, 0.505)$ for the log. **Not all barriers approach the boundary in the same way.**

The KKT conditions reappear — perturbed

Stationarity of the barrier problem:

$$\nabla f(\mathbf{x}_R) - \sum_j \frac{R}{g_j(\mathbf{x}_R)} \nabla g_j(\mathbf{x}_R) = \mathbf{0}.$$

With $\boxed{u_j(R) = \frac{R}{g_j(\mathbf{x}_R)} > 0}$:

$$\nabla f(\mathbf{x}_R) = \sum_j u_j(R) \nabla g_j(\mathbf{x}_R), \quad \boxed{u_j(R) g_j(\mathbf{x}_R) = R}.$$

In the example: $u_R = \frac{1+\sqrt{1+4R}}{2} \rightarrow u^* = 1$.

R	1	0.1	0.01	0.001
u_R	1.618	1.092	1.010	1.001

Complementary slackness

KKT (4.2): $u_j g_j = 0$.

Barrier: $u_j g_j = R \rightarrow 0$.

The barrier does not satisfy complementary slackness — it satisfies a perturbed complementarity, which tends to that of KKT as $R \rightarrow 0$ ($u_j(R) \rightarrow u_j^*$ if $\mathbf{x}_R \rightarrow \mathbf{x}^*$ and LICQ at $\mathbf{x}^*$).

And an inactive constraint?

$\min(x_1 - 1)^2 + (x_2 - 1)^2$ s.t. $9 - x_1^2 - x_2^2 \geq 0$:
 $\mathbf{x}^* = (1, 1)$, $g = 7 > 0$. With the barrier, $\mathbf{x}_R \neq (1, 1)$:
 $R = 1 \rightarrow (0.882, 0.882)$, $R = 0.1 \rightarrow (0.986, 0.986)$.

The barrier feels every constraint while $R > 0$, even inactive ones (in the exterior method, a satisfied constraint costs nothing). But $u_R = R/g \rightarrow 0 = u^*$.

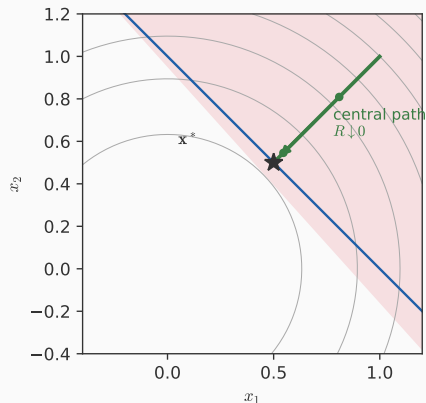
The central path

The points $\mathbf{x}_R$ obtained for $R = 1, 0.1, 0.01, \dots$ form a **path in the interior of X** that converges to $\mathbf{x}^*$ as $R \downarrow 0$ (under assumptions: f, g_j continuous; $X = \text{closure of its interior}$; $\mathbf{x}_R$ global minimizers of P , with $(\mathbf{x}_R)$ bounded; $\mathbf{x}^*$ unique — without uniqueness, every limit point is a global minimum).

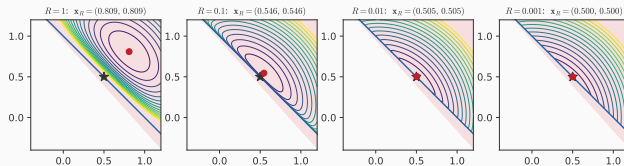
It is the **central path**.

Modern interior-point methods are based on this idea: in primal-dual formulations, they solve perturbed KKT systems ($u_j g_j = R$) and progressively reduce the barrier parameter.

In this symmetric example the path is a segment of the bisector; in general it is a curve.



The price of $R \downarrow 0$: ill-conditioning



$$\nabla^2 P = 2I + \frac{R}{g^2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \text{ and at } \mathbf{x}_R, g \approx R:$$

$$\kappa(\nabla^2 P) \approx 1 + \frac{1}{R}.$$

$R = 10^{-4}$: $\kappa \approx 10^4$. The level curves are squashed against the boundary.

The same problem, in opposite directions

Exterior: $R \uparrow \infty \Rightarrow \kappa \uparrow$. Interior:

$R \downarrow 0 \Rightarrow \kappa \uparrow$.

Getting too close to the boundary makes the subproblem hard.

The cure is the same: **a sequence of R with warm start**, never a single extreme R .

Exterior vs. interior

	Exterior (4.3.1)	Interior (log barrier)
Idea	violate and punish	prevent violation
Constraints	$g_j \geq 0$ and $h_\ell = 0$	only $g_j \geq 0$
Starting point	any	strictly feasible
Iterates	from outside	from inside
R	$\uparrow \infty$ ($c > 1$)	$\downarrow 0$ ($0 < c < 1$)
$f(\mathbf{x}_R)$	$\uparrow f^*$	$\downarrow f^*$
Error (typical)	$O(1/R)$	$O(R)$
Conditioning	$\kappa \sim R$	$\kappa \sim 1/R$
Multipliers	$u_R = -2R \langle g \rangle$	$u_R = R/g$

Interior is particularly useful when f only makes sense in X (square roots, logarithms, allowable stresses, simulators that fail outside) or when one wants to be able to stop at any moment with a feasible solution.

Exterior is particularly useful when there are equalities or no feasible point is known.

In this teaching example, with these parameters, the same inner Nelder–Mead and warm start, until $\|\mathbf{x}^{(t)} - \mathbf{x}^*\| \leq 10^{-4}$: exterior from $(0, 0)$ ($R^{(0)} = 0.1$, $c = 10$) in 6 cycles, $n_f = 844$; barrier from $(1, 1)$ ($R^{(0)} = 1$, $c = 0.1$) in 5 cycles, 743 calls to P ($n_f = 717$: 26 fall outside X); counts without the final evaluation, all cycles with tolerance 10^{-10} . $(0, 0)$ is the unconstrained minimum, the natural point for the exterior method; from the same $(1, 1)$, the exterior method takes 6 cycles, $n_f = 864$. Comparable costs, dictated by Ch. 3: each cycle is a full minimization.

Logarithmic barrier algorithm

Log barrier ($\mathbf{x}^{(0)}$ with $g_j(\mathbf{x}^{(0)}) > 0$, $R^{(0)}$, $0 < c < 1$,

$\varepsilon_{\text{comp}}$, ε_x)

1: $t \leftarrow 0$

2: **loop**

3: $\mathbf{x}^{(t+1)} \leftarrow \arg \min_{\mathbf{x}} f(\mathbf{x}) - R^{(t)} \sum_j \ln g_j(\mathbf{x})$, from $\mathbf{x}^{(t)}$
(with $P = +\infty$ if some $g_j \leq 0$: such points are rejected)

4: **if** $JR^{(t)} \leq \varepsilon_{\text{comp}}$ **and** $\|\mathbf{x}^{(t+1)} - \mathbf{x}^{(t)}\| \leq \varepsilon_x \max(1, \|\mathbf{x}^{(t)}\|)$ **then**
 return $\mathbf{x}^{(t+1)}$

5: **end if**

6: $R^{(t+1)} \leftarrow c R^{(t)}$; $t \leftarrow t + 1$

7: **end loop**

- Solve; **stay in the interior**; reduce R ; warm start.
- Practical stopping:
 $\sum_j u_j g_j = JR$ small
(perturbed complementarity)
and stalled iterates. (Note:
in convex problems, JR also
bounds $f(\mathbf{x}_R) - f^*$ from
above.)
- As a rule of thumb, moderate
 $R^{(0)}$ and $c < 1$ (in the course
examples: $R^{(0)} \approx 1$, $c = 0.1$).
- Safeguard: **never evaluate $\ln$
of a number ≤ 0** .

The price of the barrier

Advantages

- All iterates are **feasible**.
- Natural when f only exists (or only makes sense) in X .
- Estimates the multipliers: $u_j = R/g_j$.
- It is the conceptual basis of **interior-point methods**.

Limitations

- Directly handles only **inequalities**: **equalities have no interior**.
- Requires a **strictly feasible** starting point — may require a Phase I.
- The line search cannot cross the boundary.
- $R \rightarrow 0$ causes ill-conditioning ($\kappa \sim 1/R$).

When staying feasible is essential

f contains $\ln x$ or $\sqrt{x}$, or is a physical simulation that fails outside the limits — the exterior method cannot even evaluate f .

Beyond the course: primal-dual interior points (Newton on the perturbed KKT); SLP, SQP.

On the computer: MATLAB implementation

The core, in MATLAB:

```
for t = 1:tmax
    P = @(x) Pbar(f, g, x, R);           % Inf outside X
    [xn,~,~,out] = fminsearch(P, x, opt); % from x
    info.nfev = info.nfev + out.funcCount;
    u = R ./ g(xn);                     % u_j = R/g_j
    if numel(u)*R <= tolcomp && ...      % J*R = sum(u.*g)
        norm(xn - x)/max(1, norm(x)) <= tolx
        x = xn; break
    end
    x = xn; R = c*R;                     % reduce R
end
% Pbar: Inf if g_j <= 0; else f - R*sum(log(g))
```

The full function (BarrierLog, Pbar, with info.nfev and info.u) is in the course companion code.

Call:

```
f = @(x) x(1)^2 + x(2)^2;
g = @(x) x(1) + x(2) - 1;
[x, fx, info] = BarrierLog(f, g, ...
    [1 1], 1, 0.1, 1e-8, 1e-6, 12)
info % nit = 10 (R = 1e-9), u ~ 1
      % nfev ~ 140-190 per cycle
      % nfev: calls to P + 1
      % (upper bound on n_f)
```

fminsearch accepts Inf: the simplex does not accept vertices outside X (with X convex, as here; with a gradient, the line search shortens the step). funcCount counts calls to P : those at points outside X evaluate g but not f — info.nfev is therefore an upper bound on n_f .

Turbine layout of a wind farm (IDMEC, M4WIND project)

Decision the coordinates (x_i, y_i) of each of the N turbines.

Objective the annual energy produced, net of wake losses.

Constraints minimum distance between turbines — $g_{pq} = \|(x_p, y_p) - (x_q, y_q)\| - d_{\min} \geq 0$, one per pair: $J = N(N - 1)/2$ inequalities ($N = 30$: 435); site boundary and exclusion zones (geometric constraints on each turbine).

What changes vs. the textbook f is multimodal and has no reliable derivatives; KKT by hand (2^J cases) is unthinkable. The constraints typically enter by **penalty** (4.3.1) or by rejection/repair of infeasible candidates or Deb's rule (6.6), within a method from Ch. 5.

The information we have: values of f (expensive) and *geometric* constraints that are cheap to check. This asymmetry — expensive f , cheap g — decides how constraints are handled in a real problem; in the examples of this chapter f and g cost the same.

Summary table — Chapter 4

What information do we have about f and the constraints? What does it cost to obtain? What is guaranteed?

Method	Information	How it is solved / cost	Guarantee	Main limitation
Lagrange (4.1)	$\nabla f, \nabla h_\ell$ analytical	by hand: solve $n + K$ equations	necessary under independence of the ∇h_ℓ	only candidates; nonlinear system
KKT (4.2)	$\nabla f, \nabla g_j, \nabla h_\ell$	by hand: up to 2^J cases (KKT is not an algorithm)	necessary under LICQ; sufficient if convex	exponential enumeration by hand
Exterior penalty (4.3.1)	f, g_j, h_ℓ (+ gradients, if used internally)	$\sum_t n_f$ of the inner minimizations	$\mathbf{x}_R \rightarrow \mathbf{x}^*$ as $R \rightarrow \infty$ (under assumptions [†])	$\kappa \sim R$; slightly infeasible solution
Log barrier (4.3.2)	f, g_j ; interior point	idem, + Phase I	$\mathbf{x}_R \rightarrow \mathbf{x}^*$ as $R \rightarrow 0$ (under assumptions [†]); $u_j = R/g_j$	only inequalities; $\kappa \sim 1/R$

[†] f, g_j (h_ℓ) continuous; $\mathbf{x}_R$ global minimizers of P , with $(\mathbf{x}_R)$ bounded; $\mathbf{x}^*$ unique (without uniqueness: every limit point is a global minimum); for the barrier, also $X = \text{closure of the interior}$. For study. In class: the next two pages.

KEY TAKEAWAYS — 4.3.2

1. The barrier **prevents violation**: $P = f - R \sum_j \ln g_j$ only exists in the interior; it is minimized with Ch. 3 for $R \downarrow 0$.
2. It needs a **strictly feasible** point and must keep all iterates in the interior; in return, all of them are feasible.
3. The $\mathbf{x}_R$ approach $\mathbf{x}^*$ **from inside** (under assumptions), along the **central path**; typical error $O(R)$ (log).
4. The KKT conditions reappear **perturbed**: $u_j = R/g_j$, $u_j g_j = R \rightarrow 0$ — the gateway to interior-point methods.
5. The price mirrors 4.3.1: $R \downarrow 0 \Rightarrow \kappa \sim 1/R$; the cure is the same sequence with warm start. Equalities have no interior.

KEY TAKEAWAYS — Chapter 4

1. Constraints change the **directions** in which we can improve f .
2. The **multipliers** tell which constraints “count” and what they are worth.
3. **CHARACTERIZE** (4.1–4.2): Lagrange ($h = 0$: $\nabla f = \sum \lambda_\ell \nabla h_\ell$) and KKT ($g \geq 0$: $u \geq 0$, $ug = 0$) use derivatives $\rightarrow$ first-order conditions $\rightarrow$ candidates; globality in the convex case.
4. **COMPUTE** (4.3): exterior ($R \uparrow \infty$, $\mathbf{x}_R \rightarrow \mathbf{x}^*$ from outside) and interior ($R \downarrow 0$, from inside) use Ch. 3 repeatedly $\rightarrow$ cost = sum of the inner minimizations $\rightarrow$ convergence to the solution under assumptions.
5. Turning the constrained problem into unconstrained problems has a **price**: conditioning.

Next (Ch. 5): these methods give local minima. What if there are several basins?

References and resources

References Full list: *Course bibliography*, at the end of the course PDF.

- Nocedal, J., Wright, S. J., *Numerical Optimization*, 2nd ed., Springer, 2006 — Ch. 19.
- Deb, K., *Optimization for Engineering Design: Algorithms and Examples*, 2nd ed., PHI Learning, 2012.
- Deb, K., *An efficient constraint handling method for genetic algorithms*, Computer Methods in Applied Mechanics and Engineering 186(2–4):311–338, 2000 — Deb's rule.

Resources

- Companion code: BarrierLog, Pbar (`info.nfev`, `info.u`). MATLAB: `fminsearch` (accepts `Inf`).